

BPMS (Business Process Management Suite)

BPMS is an application infrastructure to support BPM projects and programs from process discovery, definition and design, to implementation, monitoring and analysis and through ongoing optimisation. BPM suites are used for the automation of business processes, to optimise business operations and to improve customer experience. BPMS can be integrated with ERP and the core line of business systems, to measure process performance and user productivity.

Some of the BPM suites worth mentioning, which are available as free software ('free' as in freedom) are featured below.

Bonita Studio CE

Bonita Studio CE is a complete free software solution for business process management and workflow, written in Java and licensed under the GNU General Public License v2. Bonita is available in both 32-bit and 64-bit versions for GNU/Linux, Windows; and in a 64-bit version for Mac platforms. It is the community edition that is *libre* and free for download. Users can graphically modify business processes using the BPMN standard, and connect the processes to other pieces of the information system to generate autonomous business applications accessible via Web portals, Web forms and mobile devices. Bonita Studio also allows the user to start with processes designed with other standards and technologies such as XPD or jBPM. It is part of the Bonita application platform with several other components – Bonita BPM Engine, Bonita Portal, Bonita UI Designer and Bonita Continuous Delivery. The latest version of Bonita Studio CE is Bonita 7.8.1 and can be downloaded from <https://www.bonitasoft.com/downloads>.

Red Hat jBPM

This is a toolkit for building business applications to help automate business processes and decisions. It provides various capabilities that simplify and

externalise business logic into reusable assets such as cases, processes, decision tables and more. It has been released under the Apache Software License. More details can be found at <https://www.jbpm.org/>.

Modelio

Developed by Modeliosoft, its core is licensed under GPL, while the Modelio Module Runtime is under the Apache Public License. Modelio combines BPMN support and UML support in one tool, with dedicated diagrams to support business process modelling. The Java Designer module uses an RCP/Eclipse-style project file structure, and supports Java code generation and reverse, Javadoc generation and Java automation. Modelio provides an XMI import/export feature that enables you to exchange UML2 models between different tools. With the HTML Publisher module, users can publish their models in HTML format. Modelio provides integrated support of the Jython scripting language, for online requests, small scripts and macro definition. It supports a wide variety of standards including UML2, BPMN2, XMI, MDA, SysML, TOGAF, SoaML and UML Testing Profile. More details can be found at <https://www.modelio.org>.

Activiti

Activiti is a cross-platform, lightweight Java based workflow engine that can automate processes written in BPMN2.0. Licensed under Apache version 2.0, the software is developed by Alfresco and the Activiti community. Activiti can be downloaded from <http://www.activiti.org/>.

Getting Started

In this article, we try out Bonita Community Edition on the GNU/Linux platform.

To ensure freedom in software usage, I suggest using any of the fully free distros like Trisquel GNU/Linux, PureOS or Debian GNU/Linux (with only the main repo enabled). Before you download Bonita Studio, make sure that you know whether you are using a 32-bit or 64-bit system, and that you have the appropriate Java version installed. In the

case of Linux, to find out whether you are using a 32-bit or 64-bit version, run the following command, which returns either 32 or 64:

```
getconf LONG_BIT
```

In my GNU/Linux system, I downloaded the latest version of Bonita using the following command:

```
wget -c https://release.ow2.org/bonita/BonitaStudioCommunity-7.8.1-x86_64.run
```

You need to change permissions of the downloaded file, which in my case is *BonitaStudioCommunity-7.8.1-x86_64.run*. To give execute permission for the software, use the following command:

```
sudo chmod +x BonitaStudioCommunity-7.8.1-x86_64.run
```

To start the installation, run the following command from the terminal:

```
./BonitaStudioCommunity-7.8.1-x86_64.run
```

Use *sudo* if the shell reports permission error. To start Bonita Studio, go to the directory where you installed it, and run the launcher for your operating system, as follows:

```
./BonitaStudioCommunity64-linux
```

In case of any trouble when running Bonita Studio, you might want to take a look at log files. Studio log file can be displayed from Bonita Studio by following the sequence *Help > Show Bonita Studio log*. If Studio fails to start, you can open the log file manually. The file is named *.log* and is located in the *<studio_folder>/workspace/.metadata* folder.

To fix any *OutOfMemory* error, you need to increase the memory allocated to the JVM that runs Bonita Studio. Edit the **.ini* file that corresponds to the executable you used to launch the Studio (e.g., *BonitaStudioSubscription64.ini* if you run *BonitaStudioSubscription64.exe*) and modify this line: *Xmx512m* to *-Xmx1024m* (or higher). Then restart Bonita Studio.